

Multi-Resolution Global Horizontal Irradiance Forecasting at Electric-Vehicle Manufacturing Locations: TimesNet and Dilated Recurrent Networks

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Abstract—This study presents an origin-aligned multi-resolution evaluation of global horizontal irradiance (GHI) forecasting at ten locations associated with electric-vehicle factories. Hourly, daily, and monthly tasks share chronological splits, forecast origins, causal GHI inputs, physical post-processing, and location-first aggregation within each resolution. The complete benchmark includes three deterministic references, XGBoost, LSTM, TimesNet, and DilatedRNN. XGBoost achieved the lowest macro-MAE in the hourly and daily tasks (50.50 W/m² at 72 hours and 43.93 W/m² at 30 days), whereas calendar climatology led the monthly tasks (12.57 W/m² at one month and 13.03 W/m² over six months). In the focused architecture comparison, TimesNet and DilatedRNN respectively obtained 52.21 and 52.96 W/m² at 72 hours, 44.08 and 43.85 W/m² at 30 days, 13.93 and 13.01 W/m² at one month, and 14.51 and 14.37 W/m² at six months. TimesNet led all hourly checkpoints and the daily comparison through 14 days; DilatedRNN led at 30 days and in the principal one-month task. Location-level wins sometimes reversed the macro-average ranking, so neither architecture was universally superior. Daily and monthly learned models use five-seed ensembles; the hourly extension uses the single registered seed 42. Claims remain descriptive because adjacent multi-step targets overlap, and meteorology is used only as independent post-hoc context.

Index Terms—Global horizontal irradiance, solar forecasting, TimesNet, DilatedRNN, multi-resolution forecasting.

I. INTRODUCTION

Pathways consistent with deep emissions reductions combine a rapid expansion of low-emission electricity with the electrification of end uses [1], [2]. Declining renewable-energy costs can reinforce this coupling [3], while scenario-based evidence indicates that the diffusion of solar photovoltaics may acquire substantial self-sustaining momentum before mid-century [4]. These system-level trends do not remove the operational difficulty created by the weather-dependent and diurnal nature of the solar resource. They instead make the temporal value of solar information increasingly important for planning, scheduling, and energy management.

The industrial sector is a particularly demanding setting for this transition. Its heterogeneous processes require combinations of efficiency, material changes, low-carbon fuels,

electrification, and low-emission electricity [5], [6]. Direct electrification can reduce emissions from technically suitable industrial heat uses, but its potential depends on process temperature, infrastructure, and the carbon intensity of the electricity supply [7]. On-site and procured renewable electricity are therefore relevant options for manufacturing, rather than a universal substitute for all industrial energy inputs [8], [9]. Adoption also faces site-specific financial, regulatory, technical, and organizational barriers [10].

Regulatory drivers must likewise be stated with their jurisdiction and conditions. For example, Directive (EU) 2024/1275 establishes a phased framework for solar-energy installations on specified building categories when technically, economically, and functionally suitable [11]. It does not apply to the Brazilian, Mexican, or United States locations examined here. Public corporate disclosures provide illustrative evidence of manufacturing-related renewable-energy adoption: Nestlé reports that its KitKat production lines in Caçapava use predominantly solar electricity [12]; Mercedes-Benz describes rooftop photovoltaics and storage at Factory 56 [13]; and Tesla and Ford report solar installations at selected manufacturing sites [14], [15]. These examples motivate the industrial context, but they are not evidence that photovoltaic systems are installed at any of the ten factory-associated coordinates used in this study.

Solar forecasting spans distinct physical and operational regimes. Intra-hour forecasts can track cloud motion, hourly and day-ahead forecasts support short-term scheduling, and forecasts aggregated over days or months address progressively longer planning scales [16], [17]. The terms short-term and long-term are consequently incomplete unless temporal resolution and forecast horizon are both reported [18], [19]. A model that is effective for a direct 72-hour sequence of hourly global horizontal irradiance (GHI) is not thereby established as effective for daily means or a one-step monthly target. Conversely, low error on an aggregated monthly series cannot be compared numerically with error on an hourly series without accounting for the different targets, sample sizes, and variability.

This article studies GHI at ten geographic points associated with electric-vehicle manufacturing facilities. The points span different latitudes and atmospheric regimes and are used as a geographically meaningful test bed. The irradiance records are satellite-derived, model-based estimates from the National

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Solar Radiation Database (NSRDB) [20]; they are not measurements from factory sensors. No factory production, electricity consumption, photovoltaic capacity, or other proprietary operational data enter the analysis. Accordingly, the forecasts characterize the solar resource at the selected grid cells and are not forecasts of factory load or photovoltaic power output.

Cabral et al. [21] investigated very-long-term GHI forecasting across 40 climatically diverse locations using a GHI-only GPT-Neo pipeline and a broad set of competitors. The present article addresses a different question and does not reuse those numerical results. It consolidates and extends two protocol-specific investigations developed by the present authors: a compact TimesNet for direct hourly forecasting and a Dilated Recurrent Neural Network (DilatedRNN) for one-step monthly forecasting. Those preliminary protocols differed in resolution, context, horizon, origins, competitors, and primary metrics. Their reported errors are retained only as development lineage; the architecture comparison in this article is rebuilt on matched samples within every task.

The research question is therefore: *how should TimesNet and DilatedRNN be compared when the available information, temporal resolution, forecast horizon, chronological origins, and evaluation rules are controlled across locations?* The article makes four contributions:

- 1) it defines a common multi-resolution formulation that distinguishes hourly multi-step, genuinely daily, and one-step monthly forecasting;
- 2) it specifies matched samples, chronological partitions, forecast origins, direct outputs, post-processing, and transparent baselines for the two architectures at every shared task;
- 3) it reports the complete benchmark before the focused architecture comparison, retaining macro, location, horizon, and seed-level evidence so that average performance cannot conceal local losses or strong simple references; and
- 4) it selects favorable, unfavorable, geographic, and meteorological cases by auditable rules, without treating ten locations as evidence of a causal climate effect or inferring rain from low GHI alone.

The remainder of the article is organized as follows. Section II critically reviews the forecasting literature and identifies the comparison gap. Section III defines GHI, temporal aggregation, forecasting horizons, TimesNet, and DilatedRNN. Subsequent sections document the data and feature engineering, the common methodology, the experimental design, and the evaluation metrics before presenting the results, limitations, and conclusions.

II. RELATED WORK

This section reviews empirical solar-forecasting studies and positions the present comparison. Definitions of irradiance, temporal support, and the two neural architectures are intentionally reserved for Section III; the purpose here is to compare what prior studies predicted, which information they used, how far ahead they forecast, and how they evaluated the forecasts.

A. Forecasting scope and evaluation practice

Solar-forecasting reviews consistently organize methods by forecast horizon, input source, target variable, and application [16], [17], [18], [22]. This organization is not merely terminological. Irradiance nowcasting from sky images, hourly GHI forecasting from lagged and meteorological variables, numerical-weather-prediction-based day-ahead forecasting, and monthly resource forecasting have different information sets and error structures. Text-mining evidence also shows that the field has diversified in both methods and applications [19]. Reviews focused on photovoltaic power reach the same general conclusion: reported performance is conditional on horizon, data, and optimization procedure [23]. GHI, solar irradiation, and photovoltaic power are therefore retained as distinct targets in this review.

Evaluation practice is equally important. Deterministic solar forecasts should be verified against explicit reference methods, with metrics selected for the decision context and reported at the relevant horizons [24]. General forecast-evaluation results caution that percentage measures and scale-dependent errors answer different questions [25]. Voyant et al. [26] further demonstrate that the appropriate naive reference depends on temporal structure and horizon; their comparison of training-free statistical references across multiple climates argues against using one convenient persistence rule as the sole benchmark in every setting.

B. From intra-hour to week-ahead forecasting

At very short horizons, cloud observations can carry information that is not contained in a univariate irradiance history. Chow et al. [27] derived cloud maps and motion vectors from 30-s total-sky images and evaluated cloud and irradiance nowcasts at a distributed test bed. Reviews of intra-hour methods distinguish such ground-imagery pipelines from satellite, numerical weather prediction, and time-series approaches [28], [29]. This literature also shows why performance during broken-cloud conditions cannot be inferred from clear-sky or annual-average behavior.

For hourly time series, hybrid networks commonly combine representation learning with recurrent dynamics. Huang et al. [30] used wavelet packet decomposition, convolution, long short-term memory (LSTM), and multilayer-perceptron branches with irradiance and meteorological inputs for 1-h-ahead prediction at three United States sites. Guariso et al. [31] compared feed-forward and LSTM networks under recursive, horizon-specific, and multi-output strategies for horizons of up to 12 h, including persistence and clear-sky references and a transfer analysis across sites. These studies differ not only in model complexity but also in whether future steps are produced recursively or jointly.

The horizon can be extended while retaining hourly outputs. Cheng et al. [32] combined one-dimensional convolution and LSTM components for day-ahead to week-ahead irradiance prediction using recent irradiance and a small set of weather variables. Zhao et al. [33] coupled improved complete ensemble empirical mode decomposition with adaptive noise

(ICEEMDAN) to TimesNet for week-ahead hourly solar irradiation forecasting. The latter establishes the relevance of TimesNet to solar forecasting, but its decomposition stage and week-ahead protocol do not isolate the effect of the TimesNet backbone or provide a matched comparison with a DilatedRNN.

C. Coarser resolution, multiple locations, and climate context

Coarser targets lead to a different learning problem. Ozoegwu [34] forecast monthly-mean daily global solar radiation with nonlinear autoregressive, exogenous-input, and hybrid neural formulations. Jung et al. [35] examined recurrent modeling for long-term photovoltaic-power forecasting, a related task whose target incorporates the conversion system and must not be equated with GHI. Cabral et al. [21] evaluated a GHI-only GPT-Neo pipeline and contemporary competitors over 40 locations and a very-long-term horizon. That work provides a cross-location precedent for long-horizon evaluation, whereas the present study addresses how two time-series architectures behave when temporal resolution and forecast design are matched.

Cross-location learning can increase the effective sample size by sharing parameters, but it also changes the estimand from a collection of independent local models to a global forecasting rule [36]. The broader recurrent-neural-network (RNN) literature emphasizes the same need to separate architectural capacity from window construction, optimization, and evaluation design [37]. Climate diversity adds another layer: Zhu et al. [38] compared several deep time-series models over horizons from minutes to 24 h, seasons, input combinations, and training-set lengths. Such factorial evaluation is informative, but climate labels and seasons remain descriptive unless the study design supports causal attribution.

Tables I–III consolidate representative primary studies in three temporal groups using factual dimensions adapted to this article’s research question. They do not rank papers by reported error because their targets, sites, horizons, inputs, and test protocols are not commensurate.

D. Research gap and positioning

Taken together, the reviewed studies supply strong architecture-specific and horizon-specific evidence, but none provides the exact matched comparison required here. The reviewed TimesNet solar application predicts week-ahead hourly irradiation after signal decomposition [33]; monthly neural studies use different targets and architectures [34]; and the very-long-term cross-location study centers on a language-model backbone [21]. Directly ordering their published errors would therefore confound model choice with temporal aggregation, information availability, location set, and evaluation protocol.

The contribution of the present study is consequently experimental rather than a claim that one architecture is universally superior. TimesNet and DilatedRNN are evaluated on common samples at each shared resolution, using the same chronological forecast origins, admissible inputs, post-processing rules, baselines, and location aggregation. Hourly multi-day forecasts

are kept distinct from forecasts of daily aggregates, and a one-step monthly forecast is not presented as a multi-month trajectory. This design directly addresses the comparability limitations identified by solar-forecast verification and benchmarking research [24], [26].

III. THEORETICAL BACKGROUND

Whereas Section II compares empirical studies, this section defines the physical target, the forecasting task, and the mechanisms of TimesNet and DilatedRNN. The distinction is important because neither an architecture name nor a nominal horizon uniquely specifies a forecasting experiment.

A. Global horizontal irradiance and temporal support

Global horizontal irradiance (GHI) is the instantaneous solar radiant flux received per unit area by a horizontal surface. For the sun above the horizon,

$$\text{GHI} = \text{DHI} + \text{DNI} \cos(\theta_z), \quad (1)$$

where DHI is diffuse horizontal irradiance, DNI is direct normal irradiance, and θ_z is the solar zenith angle [39]. Irradiance is a power density, conventionally expressed in W/m^2 ; time-integrated irradiation is an energy density and is a different quantity.

The NSRDB product used in this study estimates the radiative components from satellite observations and physical models [20]. A value at the grid cell nearest a factory coordinate is consequently a model-based estimate for that location, not a reading from a sensor mounted at the factory. This distinction also precludes interpreting GHI directly as photovoltaic power, which would require information about module area, orientation, efficiency, temperature, losses, and plant operation.

Let $y_{\ell,t}^{(1h)}$ be hourly mean GHI for location ℓ . The mean irradiance at a coarser temporal support Δ is

$$y_{\ell,\tau}^{(\Delta)} = \frac{1}{N_{\ell,\tau}} \sum_{t \in \mathcal{I}_{\ell,\tau}^{(\Delta)}} y_{\ell,t}^{(1h)}, \quad (2)$$

where $\mathcal{I}_{\ell,\tau}^{(\Delta)}$ contains the valid hourly samples in interval τ and $N_{\ell,\tau}$ is their number. Equation (2) preserves W/m^2 and defines a daily or monthly mean; it is not an accumulated daily or monthly irradiation. Coverage requirements must be applied before aggregation so that intervals with missing hours are not treated as complete.

B. Forecast resolution, context, and horizon

For a temporal resolution Δ , let $\mathbf{X}_{\ell,t} \in \mathbb{R}^{C \times P}$ contain the C observations and P admissible variables available at forecast origin t . A deterministic multi-output forecaster is a mapping

$$\hat{\mathbf{y}}_{\ell,t+1:t+H} = f_{\theta}(\mathbf{X}_{\ell,t}, \mathbf{z}_{\ell}) \in \mathbb{R}^H, \quad (3)$$

where $\mathbf{z}_{\ell}$ denotes static location information when used, $C\Delta$ is the physical context span, and $H\Delta$ is the physical forecast horizon. Thus, 72 hourly outputs and three daily outputs have the same nominal duration but different targets and

TABLE I

REPRESENTATIVE OBSERVATION-DRIVEN STUDIES AT INTRA-HOUR AND SHORT HOURLY HORIZONS. DESIGNS ARE REPORTED FACTUALLY RATHER THAN RANKED BECAUSE THEIR TARGETS AND TEST PROTOCOLS DIFFER.

Study	Target and scope	Inputs and processing	Predictor	Horizon and evaluation design
Chow et al. [27]	GHI and cloud conditions at the UC San Diego test bed	30-s total-sky images, clear-sky library, cloud map, shadows, and motion vectors	Image-based physical nowcast	Up to 5 min; six pyranometers on four partly cloudy days
Guariso et al. [31]	Hourly GHI at an Italian station, with transfer to other sites	Autoregressive irradiance windows; no decomposition	Feed-forward and LSTM networks	1–12 h; recursive, horizon-specific, and multi-output strategies; persistence and clear-sky references
Huang et al. [30]	Hourly irradiance at Denver, Clark, and Folsom	Irradiance, temperature, relative humidity, and wind speed; wavelet-packet decomposition	WPD-CNN-LSTM-MLP hybrid	1 h ahead; comparison with individual networks and climatology-persistence

TABLE II

REPRESENTATIVE DAY-AHEAD AND DIRECT MULTI-STEP SOLAR-FORECASTING STUDIES AND REFERENCES.

Study	Target and scope	Inputs and processing	Predictor	Horizon and evaluation design
Cheng et al. [32]	Hourly irradiance in Taiwan under tropical and subtropical marine conditions	Previous days of irradiance and selected temperature, humidity, and rainfall observations or forecasts	One-dimensional CNN, LSTM, skip concatenation, and dense output	Day-ahead through 7 days ahead; limited-data deployment experiment
Voyant et al. [26]	Solar-radiation series from sites in multiple climates	Little or no training history; statistical structure of the target	Five naive references (including a second-order autoregressive model) and their forecast combination	Direct multi-step benchmark evaluated with several complementary metrics
Zhao et al. [33]	Hourly solar irradiation	ICEEMDAN decomposition of the original sequence before temporal modeling	ICEEMDAN-TimesNet	Week-ahead multi-step prediction; comparison with alternative multi-step models

TABLE III

REPRESENTATIVE MONTHLY, MULTI-CLIMATE, AND VERY-LONG-HORIZON FORECASTING STUDIES.

Study	Target and scope	Inputs and processing	Predictor	Horizon and evaluation design
Ozoegwu [34]	Monthly-mean daily global solar radiation at selected locations	Radiation history, month number, and exogenous series according to formulation	NAR, NARX, and hybrid ANN	Monthly-scale forecasting; comparison among the three neural formulations
Zhu et al. [38]	Solar irradiance in a photovoltaic-greenhouse setting	Multiple meteorological-factor combinations and training-set lengths	TimesNet, Informer, Transformer, and Pyraformer	Nine horizons from 10 min to 24 h; seasonal and full-climate analyses
Cabral et al. [21]	GHI at 40 locations spanning multiple climate classes	GHI-only enhanced pattern-recognition pipeline	GPT-Neo and contemporary neural competitors	Very-long-term multi-year prediction; location and aggregate errors plus computational assessment

Abbreviations in Tables I–III: ANN, artificial neural network; CNN, convolutional neural network; GHI, global horizontal irradiance; ICEEMDAN, improved complete ensemble empirical mode decomposition with adaptive noise; LSTM, long short-term memory; MLP, multilayer perceptron; NAR, nonlinear autoregressive; NARX, nonlinear autoregressive with exogenous inputs; WPD, wavelet packet decomposition.

information content. Likewise, predicting one monthly mean is a one-step monthly forecast, not a multi-month trajectory [16], [19].

In a direct multi-output strategy, all H values in (3) are produced jointly. A recursive strategy instead feeds one-step predictions back as inputs and can accumulate error with horizon [31]. The strategy must therefore be held constant in a model comparison. A local model estimates separate parameters for each location; a global model shares θ across locations and may condition on $\mathbf{z}_\ell$ [36]. Sharing can increase the effective training sample, but it does not guarantee equal local performance.

Persistence, diurnal or seasonal naive forecasts, and simple autoregressive references expose how much skill is attributable to continuity and seasonality rather than architectural complexity. Their suitability depends on resolution and horizon, so more than one transparent reference may be needed [26].

C. TimesNet

TimesNet models one-dimensional sequences through a set of period-conditioned two-dimensional representations [40].

Let $\mathbf{X} \in \mathbb{R}^{B \times T \times C_x}$ be a batch processed by a TimesBlock, and let $F_{b,f,c}$ denote the temporal fast Fourier transform (FFT) of sample b at frequency bin f and channel c . For leakage-controlled inference, the implemented variant averages only over the channels of the same sample,

$$A_{b,f} = \frac{1}{C_x} \sum_{c=1}^{C_x} |F_{b,f,c}|. \quad (4)$$

The zero-frequency component is excluded, and each sample independently uses its k largest amplitudes to select frequency indices $f_{b,i}$ and integer periods

$$\{f_{b,1}, \dots, f_{b,k}\} = \arg \text{top-k}_{1 \leq f \leq \lfloor T/2 \rfloor} A_{b,f}, \quad p_{b,i} = \left\lfloor \frac{T}{f_{b,i}} \right\rfloor. \quad (5)$$

The retained aggregation amplitude is therefore $a_{b,i} = A_{b,f_{b,i}}$. This sample-specific selection ensures that changing other windows in a computational batch cannot change sample b 's periods.

For each $p_{b,i}$, the sequence is zero-padded to $L_{b,i} = p_{b,i} \lceil T/p_{b,i} \rceil$ and reshaped so that one axis indexes position within a period and the other indexes successive periods.

A two-dimensional Inception-style convolution then learns intraperiod and interperiod variations. If $\mathcal{G}_{b,i}$ is the processed tensor and $\mathcal{R}_{b,i}^{-1}$ restores chronological order, the period-specific outputs are combined with sample-specific amplitude-derived weights

$$\alpha_{b,i} = \frac{\exp(a_{b,i})}{\sum_{j=1}^k \exp(a_{b,j})}, \quad (6)$$

$$\tilde{\mathbf{X}}_b = \mathbf{X}_b + \sum_{i=1}^k \alpha_{b,i} \left[\mathcal{R}_{b,i}^{-1}(\mathcal{G}_{b,i}) \right]_{1:T}.$$

The residual connection preserves the original representation. Before value embedding, the implementation also standardizes each input window by its own temporal mean and population standard deviation, with 10^{-5} added to the variance, and reverses that transformation at the output. This internal, causal window standardization is applied after the external site-wise min-max transformation described in Section IV-A. Stacked TimesBlocks and a task-specific projection produce the forecast. The two-dimensional tensors are temporal reorganizations, not geographic images; their axes express within-period position and recurrence across periods. Period selection and aggregation weights are both sample-specific; batch composition is therefore a computational detail rather than part of the forecast information set. TimesNet has also been combined with signal decomposition for week-ahead hourly solar forecasting [33], but decomposition is not an intrinsic component of the TimesNet architecture.

D. Dilated recurrent neural networks

A DilatedRNN changes the recurrent connection pattern rather than reshaping the sequence [41]. Let $\mathbf{c}_t^{(q)}$ denote the recurrent state in layer q , and let $\mathbf{x}_t^{(q)}$ be the layer input, with $\mathbf{x}_t^{(1)} = \mathbf{x}_t$ and $\mathbf{x}_t^{(q)} = \mathbf{c}_t^{(q-1)}$ for $q > 1$. A dilated recurrent layer updates

$$\mathbf{c}_t^{(q)} = \phi_q \left(\mathbf{x}_t^{(q)}, \mathbf{c}_{t-d_q}^{(q)} \right), \quad (7)$$

where ϕ_q can be an ordinary RNN, gated recurrent unit, or LSTM cell [42]. The recurrent edge from $t-1$ is replaced by the edge from $t-d_q$ in that layer; dilation is therefore not an additional residual connection.

With an integer base $M > 1$, the standard exponentially increasing schedule is

$$d_q = M^{q-1}, \quad q = 1, \dots, Q. \quad (8)$$

The first layer retains local recurrence when $d_1 = 1$, while higher layers connect states at progressively coarser temporal separations. This schedule shortens recurrent paths between distant observations and exposes multiple temporal scales. It does not discard or downsample the intervening observations: the interleaved recurrent chains are recombined at their original time indices. An output head maps the resulting states to one or several forecast steps.

TimesNet and DilatedRNN consequently encode multi-scale dependence through different inductive biases. TimesNet discovers dominant Fourier periods and applies two-dimensional convolutions; DilatedRNN fixes a hierarchy of sparse recurrent lags. Neither architecture, by itself, defines the context,

forecasting strategy, horizon, or evaluation protocol. Those elements must be matched experimentally before performance differences can be attributed to the architectures.

E. Atmospheric variability and climate context

Solar geometry creates a deterministic diurnal envelope, whereas clouds and other atmospheric constituents modulate the irradiance reaching the surface. Cloud advection can create rapid GHI changes and photovoltaic power ramps [27], [43]. Aggregation reduces some high-frequency variation but does not remove persistent cloudy regimes, synoptic variability, or seasonality.

Low GHI alone is not evidence of precipitation: the value may reflect solar elevation, cloud optical properties, aerosols, or the temporal aggregation window. Descriptions such as rainy or cloudy therefore require an independent meteorological field with aligned time and location. Reanalysis products can provide such contextual fields, subject to their own spatial resolution and model uncertainty [44], [45].

Köppen–Geiger classes summarize long-term temperature and precipitation regimes [46]. They are useful for describing the geographic diversity of the sampled locations, but they are not dynamic forecast inputs unless explicitly encoded. With ten sites, differences between classes are exploratory associations rather than causal estimates of climate effects on forecast error.

IV. UNIFIED PROPOSED APPROACH

A. Data and Feature Engineering

1) *Forecasting data and spatial sampling*: The forecasting target is global horizontal irradiance (GHI), expressed as a mean irradiance in W/m^2 at each temporal resolution. The underlying product is the GOES Aggregated Physical Solar Model v4 distributed through the National Solar Radiation Database (NSRDB) [20], [47]. NSRDB irradiance is a satellite-informed, physically modelled grid product; it is not a measurement from a pyranometer installed at any factory. The factory coordinates therefore serve only to select the associated NSRDB grid cell. Throughout the article, the term *location* denotes that factory-associated grid point, not a plant-level measurement system.

The ten locations form a common panel covering Brazil, Mexico, and the United States, but the temporal tasks use two independently acquired NSRDB archives. The official hourly archive contains native 30-min records that its pipeline aggregates to one-hour means. The daily archive was requested separately at a 60-min interval and exported as arithmetic civil-day means; monthly GHI is then computed as the arithmetic mean of those daily values in each civil month. Both archives cover the same ten factory-associated locations and the nominal 2019–2024 study period, but the daily/monthly records are not reaggregations of the exact HDF5 records used by the hourly experiment. Differences caused by independent retrieval resolution or product revision are therefore part of the cross-task provenance and are one reason numerical errors are compared only within a matched task. Observations from

TABLE IV
FIELDS USED BY THE FORECASTING AND CONTEXTUAL PIPELINES.

Role	Field	Use
Target and lagged input	GHI	Direct forecast target and causal history at the task resolution.
Static model input	Site identifier	Embedding for neural models and one-hot encoding for XGBoost.
Temporal index	Timestamp	Ordering, partition boundaries, forecast origins, and target alignment; not an additional covariate.
Spatial index	Factory latitude/longitude	Selection of the NSRDB grid cell and sampling of contextual maps; not a learned input.
Learned transform	Site-wise min–max scale	Fitted strictly before the selection or test cut and reused without clipping.
Hourly physical rule	Solar elevation	Sets post-processed nighttime forecasts to zero; not an input to model fitting.
Context only	Köppen–Geiger class and NASA POWER fields	Post-hoc geographical and adverse-weather interpretation; never used for training, selection, or forecasting.

2019 provide the initial context needed by tasks whose first forecast origin occurs in 2020.

Table IV distinguishes model inputs from indexing and context-only fields. This separation is central to the matched comparison: TimesNet, DilatedRNN, LSTM, and XGBoost receive the same causal GHI history and the same site identifier. Calendar variables, coordinates, climate classes, and meteorological descriptors are not supplied as predictors.

2) *Quality control and leakage control*: Each stored source is audited according to its schema before modelling. Timestamp and GHI fields are parsed, the records are sorted chronologically, and the resulting index must be unique, monotonic, regular, and common across locations. Invalid timestamps, nonnumeric or non-finite GHI, negative GHI, duplicates, discontinuous grids, and an inconsistent stored site identity are rejected. Thus the daily loader repairs source row order by sorting rather than claiming to reject a merely unsorted input. Product and retrieval metadata are retained as provenance rather than supplied to the forecast models. Monthly aggregation is performed only after the daily checks.

For a site ℓ , the matched input at origin t is

$$\mathbf{x}_{\ell,t} = (y_{\ell,t-c}, \dots, y_{\ell,t-1}; \ell), \quad (9)$$

where c is the task-specific context length and ℓ is the fixed site identifier. Every target vector starts at t , and window construction rejects an origin if its complete target would cross the declared partition boundary. Consequently, no lag, normalization parameter, or target value is obtained from beyond the relevant cut.

Min–max parameters are estimated separately for each site. Model selection uses scales fitted to observations strictly before 1 January 2023. After the epoch count has been selected, the refitted model uses new scales fitted strictly before 1 January 2024. The same parameters transform the corresponding inputs and targets and invert the forecasts. Values outside

TABLE V
CANONICAL FACTORY COORDINATES, MAP IDENTIFIERS, AND PRESENT 1-KM KÖPPEN–GEIGER ASSIGNMENTS.

Map ID	Factory-associated location	Country	Lat., lon. (°)	Class (conf.)
1	BYD Camaçari	Brazil	−12.6734, −38.2812	Af (100%)
2	Tesla Gigafactory Nevada	USA	39.5404, −119.4391	BWk (100%)
3	Tesla Gigafactory Texas	USA	30.2219, −97.6188	Cfa (100%)
4	Hyundai Metaplant Georgia	USA	32.1605, −81.4510	Cfa (100%)
5	Rivian Normal	USA	40.5093, −89.0545	Dfa (100%)
6	Tesla Fremont Factory	USA	37.4927, −121.9415	Csb (100%)
7	Lucid AMP 1 Casa Grande	USA	32.8569, −111.7784	BWh (100%)
8	GM Factory Zero	USA	42.3815, −83.0453	Dfa (100%)
9	Ford Rouge EV Center	USA	42.3056, −83.1676	Dfa (100%)
10	BMW San Luis Potosí	Mexico	21.9683, −100.8492	BSh (50%)

Class descriptions from the distributed Beck et al. legend: Af, tropical rainforest; BWh/BWk, hot/cold arid desert; BSh, hot arid steppe; Cfa, temperate, no dry season, hot summer; Csb, temperate, dry summer, warm summer; and Dfa, cold, no dry season, hot summer. Coordinates are factory reference points; they select the associated NSRDB cell and are not sensor coordinates.

the training range are not clipped, so genuine extrapolation remains observable.

3) *Geographical and climatic context*: Table V reports the canonical factory coordinates, the map identifier used in Figure 1, and the present Köppen–Geiger class sampled at those coordinates. Classes and their confidence values come from the aligned 0.008333-degree (nominal 1-km) products of Beck et al. [46]. Sampling uses the raster cell containing each WGS84 point, without interpolation. The 0.5-degree product is used only as the background of the Americas map and never to assign a site’s class.

Nine sites have a class confidence of 100%. BMW San Luis Potosí is the exception: its BSh assignment has 50% confidence and is therefore treated as uncertain in interpretation. Climate codes, confidence values, raster-cell indices, source metadata, and file hashes are retained in the geographical context artifact described in Appendix A. They are descriptive attributes and do not alter the matched forecasting inputs in Eq. (9).

4) *Independent post-hoc meteorological context*: Daily NASA POWER fields for 2024 provide an external context layer for error interpretation [48]. These are gridded estimates rather than in-situ confirmation of weather at a factory. The retained variables are corrected precipitation, cloud amount, all-sky surface shortwave irradiation, and clear-sky surface shortwave irradiation in local solar time. The complete table contains 366 dates for each of the ten locations.

The adverse-condition rules were declared independently of forecast error: the author-defined operational tags are precipitation-indicated when corrected precipitation is at least 1 mm/day, high-cloud when cloud amount is at least 80%,

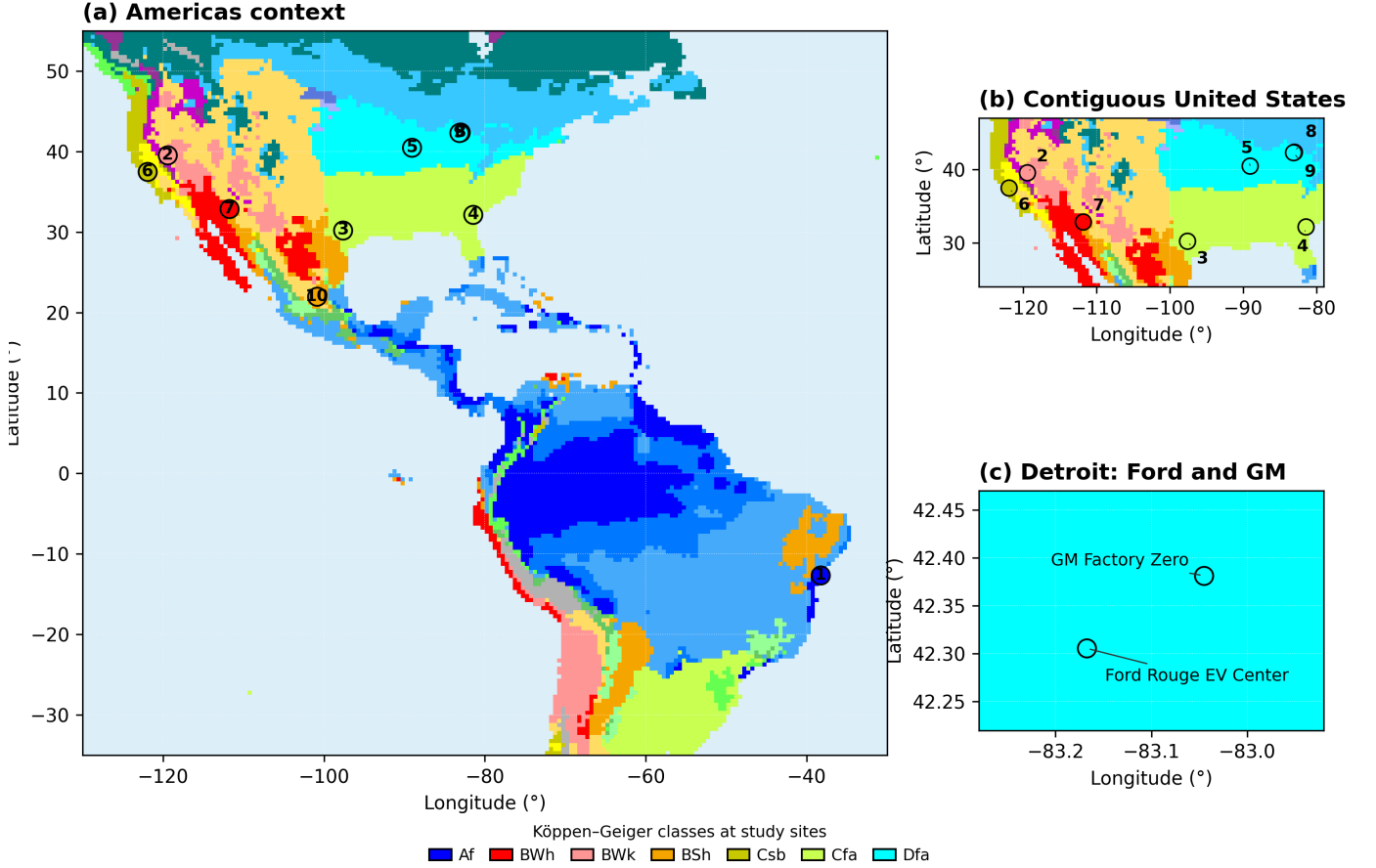


Fig. 1. Geographical and climatic context of the ten factory-associated locations. The Americas panel uses the present 0.5-degree Köppen-Geiger raster for visual context; site assignments and confidence values are sampled from the present nominal 1-km raster. Numbered markers match the Map ID column in Table V. The United States and Detroit insets separate spatially close labels. The map is descriptive and its climate fields are not forecast-model inputs. Source: Beck et al. [46].

and adverse when either condition holds. These thresholds are not presented as universal meteorological definitions and are applied only after forecasts and errors have been fixed. NASA POWER values are not joined to training, validation, refit, or test input matrices and are not used to select a model or metric. The join matches POWER local-solar-time dates to the fixed-offset NSRDB local calendar and uses factory coordinates rather than the exact NSRDB grid-cell centres. It therefore supports only an approximate, day-level, post-hoc description; it neither confirms conditions at a particular forecast hour nor supports causal attribution of forecast errors.

B. Methodology

The unified method is an origin-aligned, multi-resolution comparison rather than a numerical pooling of the two source studies. TimesNet [40] and DilatedRNN [41] are evaluated on identical windows within each matched task. The DilatedRNN source study motivates the architecture, but its quantized forecasts and previously reported errors are not imported into the present continuous-regression protocol. Figure 2 summarizes the three reproducible blocks; Algorithms 1–3 specify them separately.

1) Block A: data contract and temporal windows: Block A establishes the information set before any model is fitted. It audits the regular series, applies the task resolution, assigns a stable site identifier, creates the chronological partitions, and fits two independent sets of site-wise min-max parameters. The selection scale contains only observations before 2023, whereas the refit scale contains only observations before 2024.

The resulting sample has the input defined in Eq. (9). A timestamp determines availability and alignment but is not presented as a feature. Likewise, latitude, longitude, climate class, NASA POWER variables, and future calendar information are excluded. Holding the information set constant prevents a feature advantage from being attributed to an architecture.

2) Block B: matched model development: The deterministic references are persistence, annual seasonal naïve, and calendar climatology, with resolution-specific definitions. In the hourly task, daily-profile persistence repeats the last 24 observed local hours cyclically across the 72-hour output; the annual seasonal baseline retrieves the same local hour one year earlier. In the daily and monthly tasks, persistence repeats the most recent daily or monthly aggregate throughout the output, and the annual baseline retrieves the same civil day or month one year earlier. Every annual reference is asserted to precede

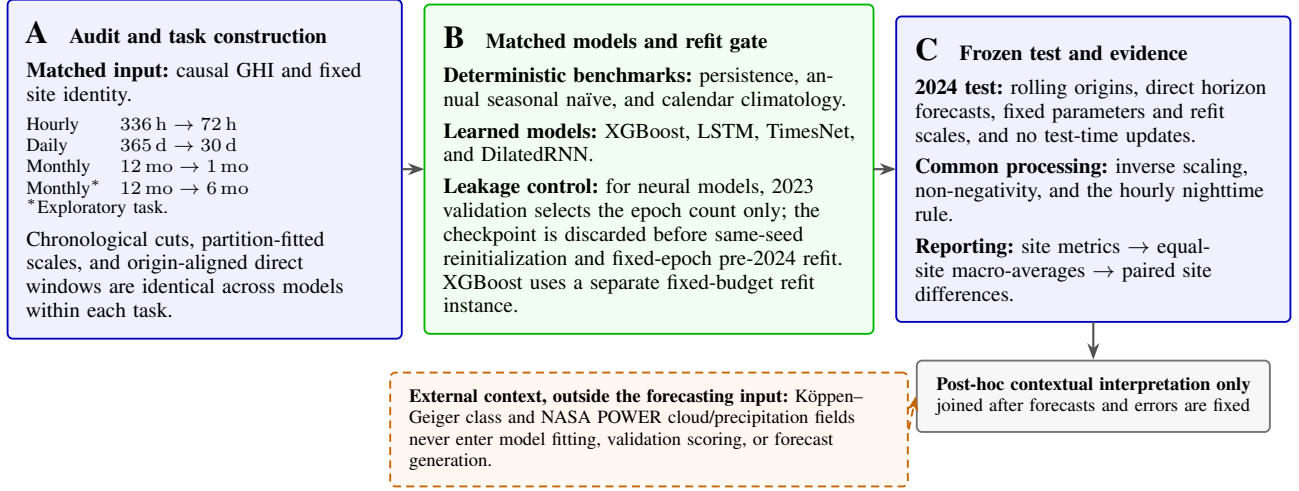


Fig. 2. Unified multi-resolution pipeline. The four frozen task contracts are evaluated on matched origins across deterministic benchmarks and learned models. External climate and weather products are excluded from model inputs and enter only after forecasts and errors are fixed. Algorithms 1–3 provide the corresponding pseudocode.

Algorithm 1 Block A: leakage-controlled data preparation

Require: Site series $\{y_{\ell,t}\}$; task $(r, c, h, \mathcal{H})$; fixed cuts
Ensure: Aligned fit, validation, refit, and test windows

- 1: **for** each site ℓ **do**
- 2: parse, sort, and verify a complete nonnegative GHI series
- 3: **if** r is monthly **then**
- 4: aggregate daily GHI by civil-month arithmetic mean
- 5: **end if**
- 6: assign the fixed identifier ℓ
- 7: fit s_{ℓ}^{sel} using observations before 2023
- 8: fit s_{ℓ}^{refit} using observations before 2024
- 9: **for** each declared partition P **do**
- 10: create (c, h) direct windows whose complete targets lie in P
- 11: assert the expected origin count and chronological boundaries
- 12: **end for**
- 13: **end for**
- 14: concatenate sites and sort each partition by origin and site identifier

the forecast origin, with 29 February mapped to 28 February. Daily and monthly climatology averages each calendar day or month using only values before the relevant cut. The official hourly artifact contains persistence and annual seasonal naïve forecasts.

The learned comparison comprises XGBoost, a direct LSTM, TimesNet, and a direct DilatedRNN. These models share the scaled GHI sequence and site identifier; neural models encode the identifier with a learned embedding, whereas XGBoost uses a one-hot representation. TimesNet selects its Fourier periods separately for every input window; consequently, batch companions and later forecast origins cannot alter a sample’s forward pass. A single global model

Algorithm 2 Block B: validation, reinitialization, and direct forecasting

Require: Block-A partitions; models $\mathcal{M}$; frozen seeds and settings
Ensure: Validation and retrospective-test forecasts on common origins

- 1: compute causal deterministic baselines at every origin
- 2: **for** each learned model $m \in \mathcal{M}$ and declared seed s **do**
- 3: **if** m is a neural model **then**
- 4: initialize m with seed s and fit on the adjustment partition
- 5: select the epoch count by 2023 normalized MSE and early stopping
- 6: retain the corresponding 2023 forecast
- 7: reinitialize m with seed s
- 8: refit on the pre-2024 partition for the selected fixed epoch count
- 9: **else**
- 10: fit fixed-configuration XGBoost on adjustment and pre-2024 windows
- 11: **end if**
- 12: generate the complete direct horizon at every 2024 origin
- 13: **end for**
- 14: retain seed-level forecasts and the daily/monthly five-seed mean

is fitted across sites, which shares temporal structure while retaining site identity [36]. Every multi-step model produces the complete target vector in one forward pass, so no predicted GHI is recursively returned as input.

Reinitialization is mandatory: the validation checkpoint is not continued into the test fit. For neural models, 2023 selects only the number of epochs for each model–seed pair. The final instance is trained from a fresh initialization on all declared pre-test windows for exactly that number of epochs.

Algorithm 3 Block C: post-processing, metrics, and contextual analysis

Require: Raw forecasts, targets, site metadata, and post-hoc context

Ensure: Site-level, macro, horizon-wise, and contextual evidence

```

1: for each model forecast do
2:   invert the scale fitted for its evaluation cut
3:   apply  $\hat{y} \leftarrow \max(\hat{y}, 0)$ 
4:   if the task is hourly then
5:     set  $\hat{y} = 0$  wherever solar elevation is nonpositive
6:   end if
7: end for
8: for each site and declared horizon do
9:   compute metrics for the cumulative prefix and the exact lead
10: end for
11: average site metrics with equal site weight
12: form TimesNet–DilatedRNN differences paired by site
13: attach climate and adverse-day labels only after errors are fixed
  
```

XGBoost has a fixed tree budget and is independently fitted on the adjustment and refit partitions. The hourly extension is narrower: all official model forecasts are read without modification, and only DilatedRNN with seed 42 is added and aligned by site, origin, target timestamp, and lead.

All partitions are evaluated in rolling-origin form. Model parameters and the refit scale remain fixed throughout the 2024 test, but the causal context is advanced at every eligible origin and may include GHI observations from earlier in 2024 once they have become historical. No test target enters a context before its timestamp, and no weights are updated during the test. The design is thus a retrospective walk-forward evaluation with fixed parameters, not a single forecast issued at the end of 2023 for every target in 2024.

3) *Block C: physical consistency and evaluation:* After inverse scaling, all forecasts receive the same non-negativity floor. Daily and monthly forecasts have no upper truncation. Hourly forecasts also use solar elevation computed at the centre of each hourly interval; predictions are set to zero when elevation is nonpositive, following the deterministic geometry implemented with `pvl` [49]. Solar elevation is therefore a post-processing constraint, not privileged predictive information.

The unit of the matched descriptive comparison is the location. For each metric and horizon scope, the pipeline retains the TimesNet–DilatedRNN difference, its mean and median across the ten sites, and the number of sites won by TimesNet. It does not treat overlapping forecast origins as independent replicates. The NASA POWER join occurs only in the final interpretive layer, using the predeclared operational 1 mm/day and 80% thresholds from Section IV-A; it cannot influence training, model selection, or the reported aggregate metrics.

C. Experimental Setup

The executable study defines four frozen tasks rather than comparing errors across unlike resolutions. Table VI lists their contexts, direct outputs, evaluation horizons, and expected origin counts. Within a task, all models share sites, origins, targets, preprocessing cuts, and post-processing. Across tasks, metrics retain their resolution-specific units and are not pooled.

1) *Common matched protocol:* The daily and monthly experiments contain three deterministic references (persistence, annual seasonal naïve, and calendar climatology) and four learned references (XGBoost, LSTM, TimesNet, and DilatedRNN). Every learned model receives only causal GHI and site identity. Neural models use the five seeds 11, 23, 42, 67, and 89, and XGBoost is likewise refitted under each of those seeds because row and column subsampling are stochastic; both seed-level forecasts and the arithmetic mean of every learned model are retained. Deterministic references are evaluated once. The primary model comparison uses macro-MAE after computing MAE separately at each site. RMSE, nRMSE, and R^2 , as well as cumulative and exact-lead scopes, remain mandatory complementary outputs.

For selection, neural networks are initialized under each seed, fitted to the adjustment windows, and monitored on the 2023 windows using normalized MSE. Early stopping chooses an epoch count for each model–seed pair. The network is then discarded, reinitialized under the same seed, and fitted to the pre-2024 refit windows for exactly that number of epochs. XGBoost uses its frozen tree budget in separate adjustment and refit instances. Test forecasts are generated only by the refitted instances.

Figure 3 summarizes this selection–refit chronology and the four direct-output contracts that remain fixed during testing.

Within each partition, forecasts follow the rolling-origin protocol of Section IV-B: the official hourly task advances one local- midnight origin per day, the daily task advances one origin per day, and the monthly tasks advance one origin per civil month. Parameters remain fixed in the test partition while each context incorporates only observations that have become available before its first target.

2) *Hourly protocol and DilatedRNN extension:* The official hourly task uses 336 past hours (14 days) and produces the next 72 hours directly. The reported checkpoints are 24, 48, and 72 hours; they are prefixes and exact leads extracted from the same 72-hour output, not separately trained models. Adjustment origins run from 15 January 2019 through 29 December 2022 (1,445 per site), and validation origins run from 1 January through 29 December 2023 (363 per site). The pre-test refit combines origins from 15 January 2019 through 29 December 2023 (1,810 per site). The retrospective test contains 363 origins from 1 January through 28 December 2024. The last 72-hour window ends at 23:00 on 30 December in the fixed local-time index; corresponding UTC timestamps can extend into 31 December for western locations.

The official artifact fixes seed 42 and contains persistence, annual seasonal naïve, XGBoost, LSTM, and TimesNet predictions. The extension trains only a direct DilatedRNN with seed 42 under the same 336-to-72 contract. Its forecasts are merged one-to-one with the read-only official artifact by site,

TABLE VI
FROZEN MULTI-RESOLUTION FORECASTING TASKS.

Task	Context $\rightarrow$ output	Evaluated horizons	Origins/site A/V/R/T	Role
Hourly	336 h $\rightarrow$ 72 h	24, 48, 72 h	1,445/363/1,810/363	Official protocol; DilatedRNN seed-42 extension
Daily	365 d $\rightarrow$ 30 d	1, 3, 7, 14, 30 d	1,067/336/1,432/337	Primary matched task
Monthly-1	12 mo $\rightarrow$ 1 mo	1 mo	36/12/48/12	Primary monthly task
Monthly-6	12 mo $\rightarrow$ 6 mo	1, 3, 6 mo	31/7/43/7	Exploratory multi-month task

A/V/R/T denote adjustment, 2023 validation, pre-2024 refit, and 2024 test. Every count is the number of complete forecast origins per site. Reported multi-step horizons are evaluated both as cumulative prefixes and exact leads.

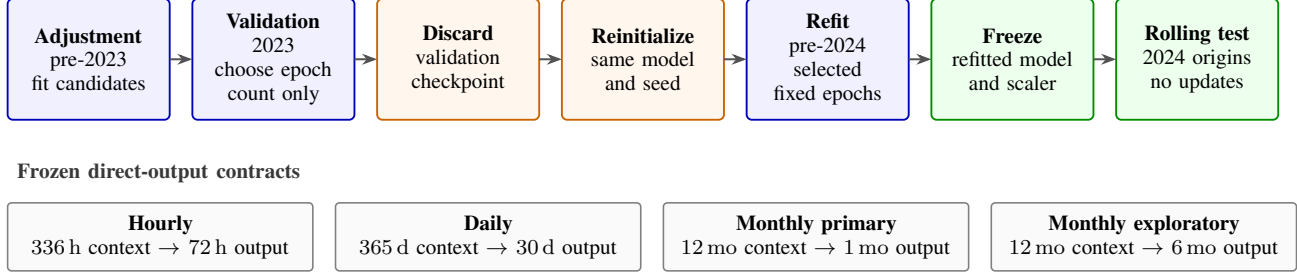


Fig. 3. Temporal selection, refit, and evaluation protocol for each neural model–seed pair. Blue boxes fit or score on historical partitions, orange boxes enforce fresh retraining, and green boxes denote the locked test state. XGBoost follows the analogous separation with a fixed tree budget and independently fitted adjustment and refit instances. Each rolling origin emits the complete direct horizon shown in the lower band.

forecast origin, target timestamp, and lead. Target equality is checked during the one-to-one merge, and the hashes of all official inputs are rechecked before the extension is marked complete. Thus the hourly extension does not silently retrain or overwrite an official comparator.

3) *Daily protocol*: The daily primary task uses 365 daily means to predict a 30-day vector. Metrics are retained at 1, 3, 7, 14, and 30 days. Adjustment origins extend from 1 January 2020 through 2 December 2022, with targets ending no later than 31 December 2022 (1,067 origins per site). Validation uses 1 January–2 December 2023 (336 origins), and refit uses 1 January 2020–2 December 2023 (1,432 origins). The 2024 test uses 1 January–2 December and contains 337 complete origins per site because 2024 is a leap year.

4) *Monthly primary and exploratory protocols*: Civil-month means are calculated from the audited daily series. The primary monthly task uses 12 months to predict the next month. It has 36 adjustment origins in 2020–2022, 12 validation origins in 2023, 48 refit origins in 2020–2023, and 12 test origins in 2024, for every site.

The exploratory monthly task uses the same 12-month context and directly predicts six months. Its declared checkpoints are months 1, 3, and 6. Complete-target constraints restrict origins to January–July: adjustment has 31 origins from January 2020 through July 2022, validation has seven in 2023, refit has 43 through July 2023, and test has seven in 2024. This task is explicitly labelled exploratory because each site contributes only seven test origins and the output covers one half-year.

For the six-month task, exact leads are evaluated on different target-month sets: lead 1 covers January–July, whereas lead 6 covers June–December. Its horizon profile therefore combines

TABLE VII
COMPACT FROZEN CONFIGURATION OF THE MATCHED IMPLEMENTATION.

Component	Frozen setting
Matched input	Causal GHI sequence plus fixed site identifier only
Regression/output	Continuous GHI; direct vector of task-specific length
Neural optimization	Adam; MSE loss; learning rate 10^{-3} ; weight decay 10^{-5}
Batch size	128 for hourly/daily; 32 for monthly
Epoch selection	Hourly/daily: maximum 30, patience 5; monthly: maximum 300, patience 30
Seeds	Daily/monthly: 11, 23, 42, 67, 89; official hourly and DilatedRNN extension: 42
TimesNet	$d_{\text{model}} = 8$, $d_{\text{ff}} = 16$, one TimesBlock, top- $k = 3$, two kernels, dropout 0.1
DilatedRNN	Dilations (1, 2, 4); 16 units/layer; dense width 16; dropout 0
LSTM	One layer; hidden width 32; direct output head
Site embedding	Width 8 for all neural models
XGBoost	150 trees; depth 3; learning rate 0.05; row and column subsampling 0.8; histogram tree method
Post-processing	Zero floor for all tasks; hourly values also set to zero when solar elevation is nonpositive

lead-time and seasonal effects and is not interpreted as an isolated degradation curve.

5) *Frozen optimization settings*: Table VII gives the compact configuration needed to interpret the main comparison. Appendix A records the full temporal boundaries, model settings, baseline definitions, artifact contract, and external-context integrity hashes.

6) *Model capacity and computational evidence*: Following the capacity reporting used to interpret architecture compar-

isons, Table VIII gives the exact number of trainable scalar parameters in each neural model. The counts are obtained from the frozen module dimensions, including biases and the tensite embedding, rather than inferred from checkpoint file sizes. TimesNet has a task-dependent direct temporal projection, so its parameter count grows with both context and output length. Dilated recurrence changes the temporal connections without adding separate weights for each skipped position. XGBoost is instead constrained by 150 trees of maximum depth three and is not assigned an artificial neural-parameter equivalent.

Training and inference times were not instrumented consistently in the completed runs. They are therefore not reconstructed from file timestamps, epoch counts, hardware assumptions, or checkpoint sizes. The comparison reports exact architectural capacity and the common one-thread execution policy, while treating measured runtime as unavailable evidence rather than a quantity to be estimated retrospectively.

D. Evaluation Metrics

Let y_i be a reference GHI value, $\hat{y}_i$ its post-processed forecast, $e_i = y_i - \hat{y}_i$, $\bar{y}$ the mean reference value in the same evaluation vector, and n its number of target points. Each metric is computed first for one site, one model estimate, one partition, one declared horizon, and one horizon scope.

1) Mean absolute error:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|. \quad (10)$$

MAE is the primary error measure and remains in W/m^2 . Its linear penalty makes the magnitude directly interpretable and less dominated by individual large deviations than a squared-error score [25]. Lower values are better.

2) Root mean squared error:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2}. \quad (11)$$

RMSE is also expressed in W/m^2 , but squaring gives larger deviations more weight. Reporting it alongside MAE reveals whether occasional large errors change the model ordering, a standard concern in deterministic solar-forecast verification [24]. Lower values are better.

3) Normalized root mean squared error:

$$\text{nRMSE} = \frac{\text{RMSE}}{\bar{y}}, \quad \text{nRMSE}_{\%} = 100 \text{ nRMSE}. \quad (12)$$

The machine-readable artifacts store the dimensionless ratio; tables that use percent explicitly multiply it by 100. Normalization supports comparison among sites with different mean irradiance levels, but it inherits the instability of a small denominator [26]. The value is therefore left undefined when $\bar{y}$ is numerically zero. Lower values are better.

4) Coefficient of determination:

$$R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (13)$$

R^2 measures variation tracked relative to a constant predictor built from the evaluation mean [24]. It is undefined for

fewer than two targets or a constant reference vector and can be negative when the forecast is worse than that constant reference. Higher values are better; R^2 does not replace the scale-dependent errors.

5) *Horizon scopes and spatial aggregation:* For a declared horizon H , the cumulative scope concatenates leads $1, \dots, H$ over all valid origins, whereas the exact-lead scope retains only lead H . The two scopes are identical at $H = 1$. This distinction is applied at 24, 48, and 72 hours; 1, 3, 7, 14, and 30 days; and, for the exploratory multi-month task, 1, 3, and 6 months. A horizon label therefore always denotes both a numerical lead and an explicitly named scope.

If M_ℓ is any valid site-level metric and $L = 10$, the macro result is

$$M_{\text{macro}} = \frac{1}{L} \sum_{\ell=1}^L M_\ell. \quad (14)$$

Equal site weighting prevents a site with more valid target points from dominating the geographical summary. Macro-MAE is the predeclared primary metric at every resolution; MAE, RMSE, nRMSE, and R^2 are nevertheless retained together rather than selectively reported.

For daily and monthly learned models, seed-level metrics quantify variability, whereas the TimesNet–DilatedRNN paired comparison uses the arithmetic five-seed ensemble forecast. The hourly extension compares the single registered seed 42 for both models. For each site, metric, horizon, and scope, the pipeline stores

$$\Delta_{\ell,M} = M_{\ell,\text{TimesNet}} - M_{\ell,\text{DilatedRNN}}. \quad (15)$$

A negative $\Delta_{\ell,M}$ is a TimesNet win for the three error metrics; a positive value is a win for R^2 . Mean and median differences and the number of site wins are descriptive. Overlapping multi-step windows are not treated as independent observations for an i.i.d. significance test.

V. RESULTS AND DISCUSSION

A. Matched multi-resolution benchmark

Table IX reports the equal-site-weighted macro-MAE for every cumulative checkpoint in the matched protocol. XGBoost ranked first at every daily and hourly checkpoint, with 30-day and 72-hour MAEs of 43.93 and 50.50 W/m^2 , respectively. Calendar climatology ranked first in both monthly tasks, reaching 12.57 W/m^2 at one month and 13.03 W/m^2 over the six-month prefix. These baseline results show that the architecture comparison should not be read as a claim that either neural model dominates the complete benchmark set.

The strongest result is therefore not a universal advantage of either neural architecture. Under the controlled GHI-only information set, the fixed XGBoost configuration provided the best fine-resolution benchmark, while the calendar mean remained difficult to improve upon with the short monthly history. These outcomes do not establish why a method won, but they show that architectural sophistication alone did not determine forecast skill. They also justify retaining the complete benchmark before examining the narrower TimesNet–DilatedRNN contrast.

TABLE VIII

TRAINABLE SCALAR PARAMETERS OF THE NEURAL MODELS UNDER EACH DIRECT FORECASTING TASK. COUNTS INCLUDE BIASES AND THE TEN-SITE EMBEDDING; FIXED BUFFERS ARE EXCLUDED.

Task	Context–output	LSTM	TimesNet	DilatedRNN
Hourly	336–72	7,512	27,009	3,096
Daily	365–30	5,790	13,725	2,382
Monthly primary	12–1	4,601	2,758	1,889
Monthly exploratory	12–6	4,806	2,823	1,974

TABLE IX

MACRO TEST MAE BY TASK AND CUMULATIVE FORECAST HORIZON. COMPLETE METRICS AND EXACT-LEAD RESULTS ARE PROVIDED IN THE CSV ARTIFACT.

Task	Horizon	Climatology	DilatedRNN	LSTM	Persistence	Seasonal naive	TimesNet	XGBoost	Seeds
Hourly 336–72	24	NA	50.73	50.91	52.58	65.91	49.08	48.37	1
Hourly 336–72	48	NA	52.15	52.79	56.16	65.92	51.86	49.81	1
Hourly 336–72	72	NA	52.96	53.39	58.11	65.90	52.21	50.50	1
Daily 365–30	1	45.74	43.27	42.04	43.87	58.90	43.20	41.58	5
Daily 365–30	3	45.76	43.80	43.14	50.17	58.98	43.63	43.09	5
Daily 365–30	7	45.75	44.08	43.69	53.77	59.00	43.95	43.71	5
Daily 365–30	14	45.64	44.11	44.00	56.41	58.91	44.09	43.96	5
Daily 365–30	30	45.51	43.85	44.18	60.28	58.78	44.08	43.93	5
Monthly 12–1	1	12.57	13.01	14.27	35.09	16.94	13.93	14.82	5
Monthly 12–6	1	13.61	14.89	15.74	33.19	16.50	14.88	14.73	5
Monthly 12–6	3	13.08	14.33	15.33	61.45	17.06	14.56	14.93	5
Monthly 12–6	6	13.03	14.37	14.76	91.89	18.00	14.51	14.72	5

Figure 4 summarizes the rank trajectory across all reported cumulative checkpoints of each task. Results are not pooled across resolutions because an hourly, daily, and monthly target represents a different aggregation and operational horizon.

B. TimesNet–DilatedRNN comparison

Table X gives the site-paired comparison; a negative mean or median difference denotes lower TimesNet MAE. TimesNet had the lower macro-MAE at all hourly checkpoints, by 1.65, 0.29, and 0.75 W/m² at 24, 48, and 72 hours. Its site wins nevertheless fell from eight at 24 hours to four at both 48 and 72 hours, so a minority of larger TimesNet gains determined the later macro averages.

In the daily task, TimesNet led by 0.07–0.17 W/m² through seven days and by 0.02 W/m² at 14 days. At 30 days, DilatedRNN led by 0.23 W/m² and won six sites. DilatedRNN also led the principal one-month task by 0.93 W/m² and won six sites. In the exploratory six-month task, the models were effectively tied at its one-month prefix (a 0.005 W/m² TimesNet advantage and five wins each); DilatedRNN led by 0.23 and 0.14 W/m² at three and six months and won seven sites at the terminal horizon. These changes in sign and win count rule out a universal architecture ranking.

Figure 5 retains the ten site-level terminal comparisons as signed MAE differences rather than reporting only aggregate wins. The mixed signs expose both geographic heterogeneity and the occasions on which a task-level winner loses locally.

C. Variability across seeds

Daily and monthly estimates use five independent seeds. As shown in Table XI, the 30-day seed-level MAE was

44.29 ± 0.29 W/m² for TimesNet and 44.10 ± 0.60 W/m² for DilatedRNN. At one month it was 14.61 ± 0.76 versus 13.45 ± 0.35 W/m², respectively. The exploratory six-month values were 15.29 ± 0.81 and 14.77 ± 0.45 W/m². The hourly rows contain one seed and therefore do not estimate seed variability. Table XI and Figure 6 summarize the mean and one standard deviation of seed-level metrics. These values need not equal Table IX, which evaluates the arithmetic ensemble forecast rather than averaging the nonlinear seed-level metric.

The scale of the architecture differences must be read together with this variability. At 30 days, the 0.23 W/m² ensemble advantage of DilatedRNN is smaller than either model’s seed-level standard deviation. The one-month advantage of 0.93 W/m² is larger, but still accompanies overlapping seed-level ranges. At six months, the 0.14 W/m² terminal difference is again small relative to seed variability. These comparisons are descriptive rather than confidence intervals, but they prevent small changes in rank from being presented as stable architecture dominance.

D. Relation to the source protocols

The protocol-specific studies that motivated the unified design remain useful as lineage evidence, but not as a head-to-head comparison. In the hourly source protocol, TimesNet achieved a macro-RMSE of 101.04 W/m², compared with 104.68 W/m² for the LSTM, a relative reduction of 3.48%. It also obtained lower RMSE than the LSTM at the 24- and 48-hour prefixes and lower RMSE than both temporal references at all three prefixes.

In the one-step monthly source protocol, the consolidated DilatedRNN forecast achieved a macro-MAE of 13.42 W/m²,

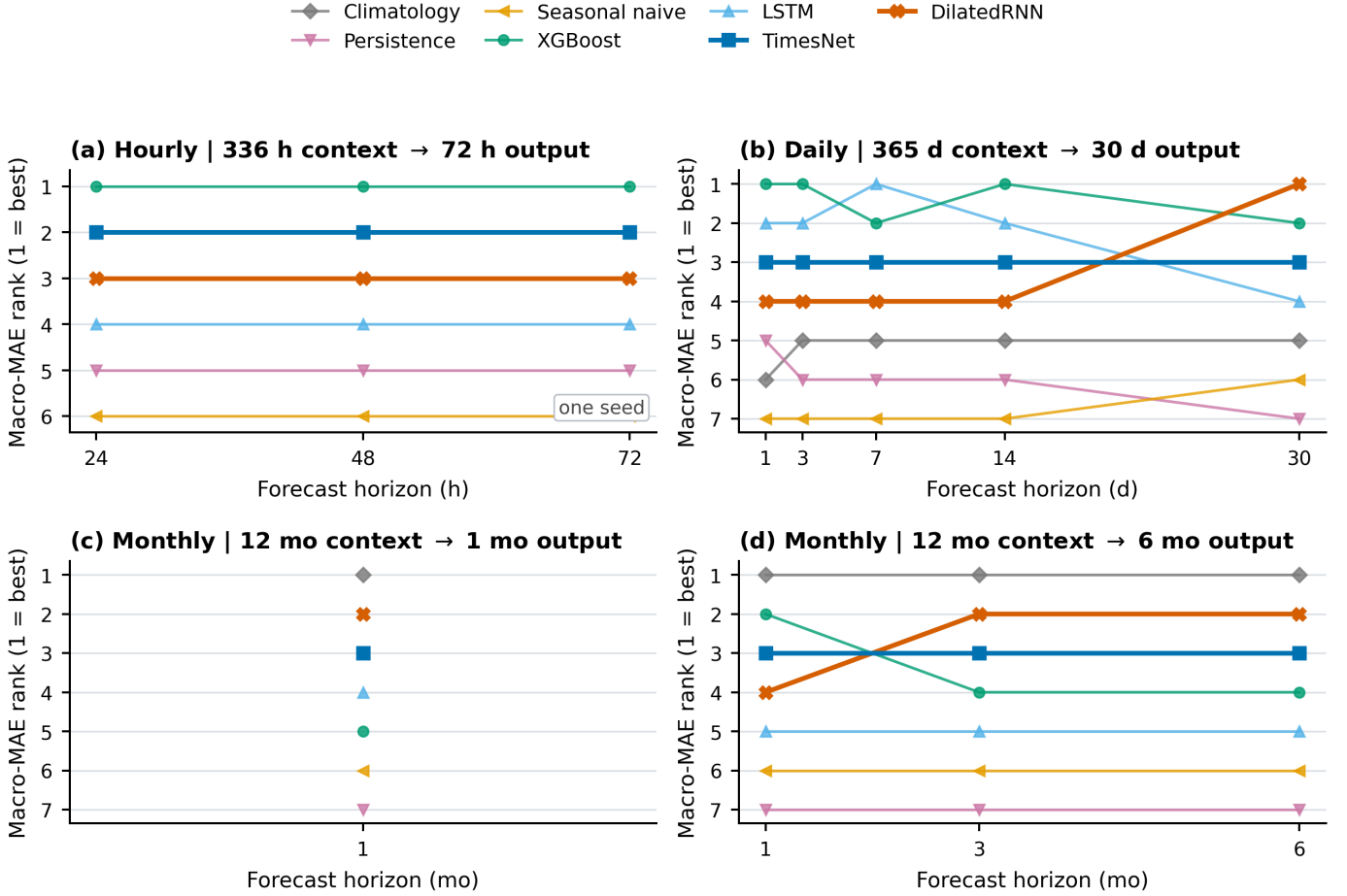


Fig. 4. Macro-MAE rank trajectories at the reported cumulative checkpoints. Rank 1 is best. Panel titles state each input–output contract, and axis units follow its native resolution. Colors, markers, and line weight jointly identify models; connecting lines only guide the eye. Ranks are not pooled across tasks.

an RMSE of 16.20 W/m^2 , and an R^2 of 0.898. It ranked first among five evaluated methods by macro-MAE but led at only four of ten locations, with local MAE from 7.45 to 20.92 W/m^2 . These two source results use different targets, inputs, competitors, and evaluation rules. Their numerical values are consequently reported separately and do not determine the matched rankings above.

E. Favorable and unfavorable cases

The global case rule was fixed before meteorological annotation: for every site, origin, and cumulative horizon, it selects the maximum and minimum of $\text{MAE}_{\text{DilatedRNN}} - \text{MAE}_{\text{TimesNet}}$. The strongest TimesNet gain occurred for GM Factory Zero at the 24-hour forecast from 31 March: 46.30 versus 109.90 W/m^2 , a 63.61 W/m^2 advantage. Post-hoc NASA POWER context described 99.64% mean cloud amount and 0.80 mm precipitation. The strongest deficit occurred for Tesla Gigafactory Texas at the 24-hour forecast from 28 March: 150.82 versus 73.92 W/m^2 , a 76.90 W/m^2 TimesNet disadvantage, with no independently flagged adverse day. Both are hourly single-seed cases and illustrate why selected examples cannot replace the full aggregate evaluation. Figure 7 retains the model MAEs, signed advantage, independent weather

annotation, and the corresponding forecast trajectories in one view.

F. Geographic and climatic interpretation

Figure 1 locates the ten sites in their descriptive Köppen–Geiger context. With only ten purposively selected locations, apparent differences between zones cannot establish a causal climatic effect. Sites in the same country may remain far apart and occupy distinct regimes, while nearby sites may share correlated weather.

As an error-independent check, the predeclared meteorological rule selected Hyundai Metaplant Georgia on 5 August 2024, with 82.66 mm/day corrected precipitation and 99.47% cloud amount. This case was selected without consulting either model’s error and is descriptive rather than causal. Its complete context record is reported in Table XII.

G. Threats to Validity and Limitations

1) *Construct and data validity*: NSRDB GHI is a satellite-informed physical-model estimate [20], not an on-site pyranometer record. Factory coordinates select product grid cells, so neither the target nor its error quantifies the irradiance at a particular roof surface. The purposive set of ten factory-associated locations is also not a sample of factory electricity

TABLE X
DESCRIPTIVE PAIRED COMPARISON OF TIMESNET AND DILATEDRNN ACROSS SITES. OVERLAPPING TARGET WINDOWS ARE NOT TREATED AS INDEPENDENT OBSERVATIONS.

Task	Horizon	Sites	Mean MAE difference (W/m ²)	Median MAE difference (W/m ²)	TimesNet site wins	DilatedRNN site wins	Ties
Daily 365–30	1	10	-0.07	-0.04	5	5	0
Daily 365–30	3	10	-0.17	-0.15	5	5	0
Daily 365–30	7	10	-0.13	-0.23	6	4	0
Daily 365–30	14	10	-0.02	-0.09	6	4	0
Daily 365–30	30	10	0.23	0.09	4	6	0
Hourly 336–72	24	10	-1.65	-1.71	8	2	0
Hourly 336–72	48	10	-0.29	0.25	4	6	0
Hourly 336–72	72	10	-0.75	0.16	4	6	0
Monthly 12–1	1	10	0.93	0.55	4	6	0
Monthly 12–6	1	10	-0.00	0.05	5	5	0
Monthly 12–6	3	10	0.23	0.12	4	6	0
Monthly 12–6	6	10	0.14	0.85	3	7	0

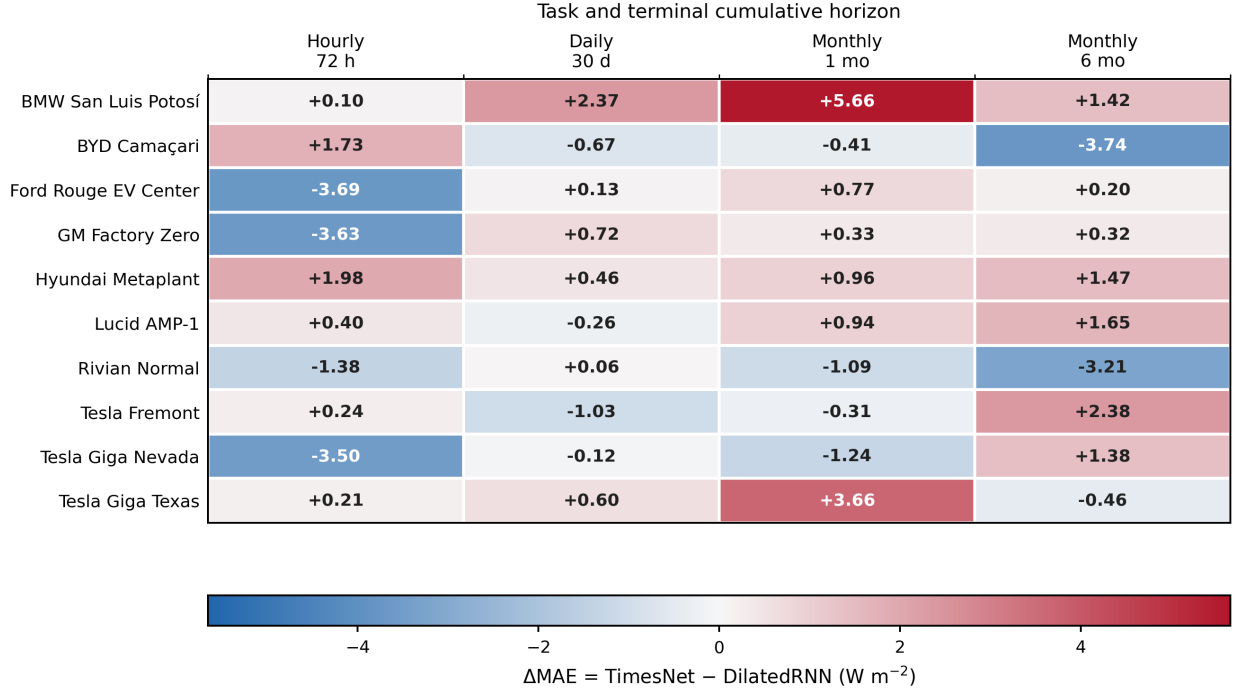


Fig. 5. Site-level terminal cumulative contrast $\Delta\text{MAE} = \text{MAE}_{\text{TimesNet}} - \text{MAE}_{\text{DilatedRNN}}$ in W/m². Annotated negative (blue) cells favor TimesNet; positive (red) cells favor DilatedRNN. Columns correspond to 72 h, 30 d, 1 month, and 6 months. The comparison is descriptive because overlapping target windows are not treated as independent observations.

demand, photovoltaic output, roof area, or operational constraints. The industrial framing therefore identifies geographically relevant forecasting sites; it does not establish plant-level energy savings.

The matched input deliberately contains only causal GHI and site identity. This restriction isolates architectural effects, but it omits potentially useful exogenous forecasts such as cloud motion, numerical weather prediction, aerosols, and temperature. A site embedding can also encode persistent location-specific behaviour and does not demonstrate transfer to an unseen factory. Fixed local offsets produce regular hourly grids but do not reproduce daylight-saving clock changes.

2) *Temporal and statistical validity*: All test evidence is retrospective and restricted to 2024. It does not measure prospective robustness under later satellite-product revisions, climate nonstationarity, or new sites. The monthly primary task

has only 48 pre-test refit origins per site, and the exploratory six-month task has seven test origins per site. Its long output therefore cannot support the same empirical resolution as the hourly or daily tasks.

Targets from adjacent multi-step origins overlap, creating strongly dependent errors. Site-first aggregation avoids allowing that overlap to change spatial weights, but it does not make the origins independent. The paired summaries accordingly use ten locations as descriptive units and do not attach an i.i.d. significance test to thousands of overlapping target points.

3) *Model-comparison validity*: TimesNet and DilatedRNN share inputs, origins, seeds, optimization limits, and post-processing in the daily and monthly tasks, but their parameterizations and inductive biases are not identical. The frozen configuration is a controlled comparison rather than an exhaustive architecture-specific search. Moreover, the hourly extension

TABLE XI
MACRO TEST MAE ACROSS INDEPENDENT SEEDS (MEAN AND STANDARD DEVIATION). SINGLE-SEED EXTENSIONS ARE IDENTIFIED EXPLICITLY.

Task	Horizon	DilatedRNN	LSTM	TimesNet	XGBoost
Hourly 336–72	24	50.73 (n=1)	50.91 (n=1)	49.08 (n=1)	48.37 (n=1)
Hourly 336–72	48	52.15 (n=1)	52.79 (n=1)	51.86 (n=1)	49.81 (n=1)
Hourly 336–72	72	52.96 (n=1)	53.39 (n=1)	52.21 (n=1)	50.50 (n=1)
Daily 365–30	1	43.51 (0.68)	42.31 (0.55)	43.55 (0.85)	41.67 (0.12)
Daily 365–30	3	44.02 (0.62)	43.45 (0.48)	43.89 (0.54)	43.16 (0.07)
Daily 365–30	7	44.29 (0.57)	44.05 (0.36)	44.17 (0.37)	43.78 (0.06)
Daily 365–30	14	44.33 (0.60)	44.42 (0.58)	44.29 (0.29)	44.02 (0.03)
Daily 365–30	30	44.10 (0.60)	44.68 (0.71)	44.29 (0.29)	44.00 (0.02)
Monthly 12–1	1	13.45 (0.35)	14.51 (0.34)	14.61 (0.76)	14.86 (0.17)
Monthly 12–6	1	15.24 (0.51)	16.25 (0.65)	15.81 (1.67)	14.83 (0.31)
Monthly 12–6	3	14.72 (0.49)	15.75 (0.24)	15.29 (0.87)	14.99 (0.14)
Monthly 12–6	6	14.77 (0.45)	15.14 (0.36)	15.29 (0.81)	14.80 (0.11)

—○— XGBoost —△— LSTM —□— TimesNet —×— DilatedRNN

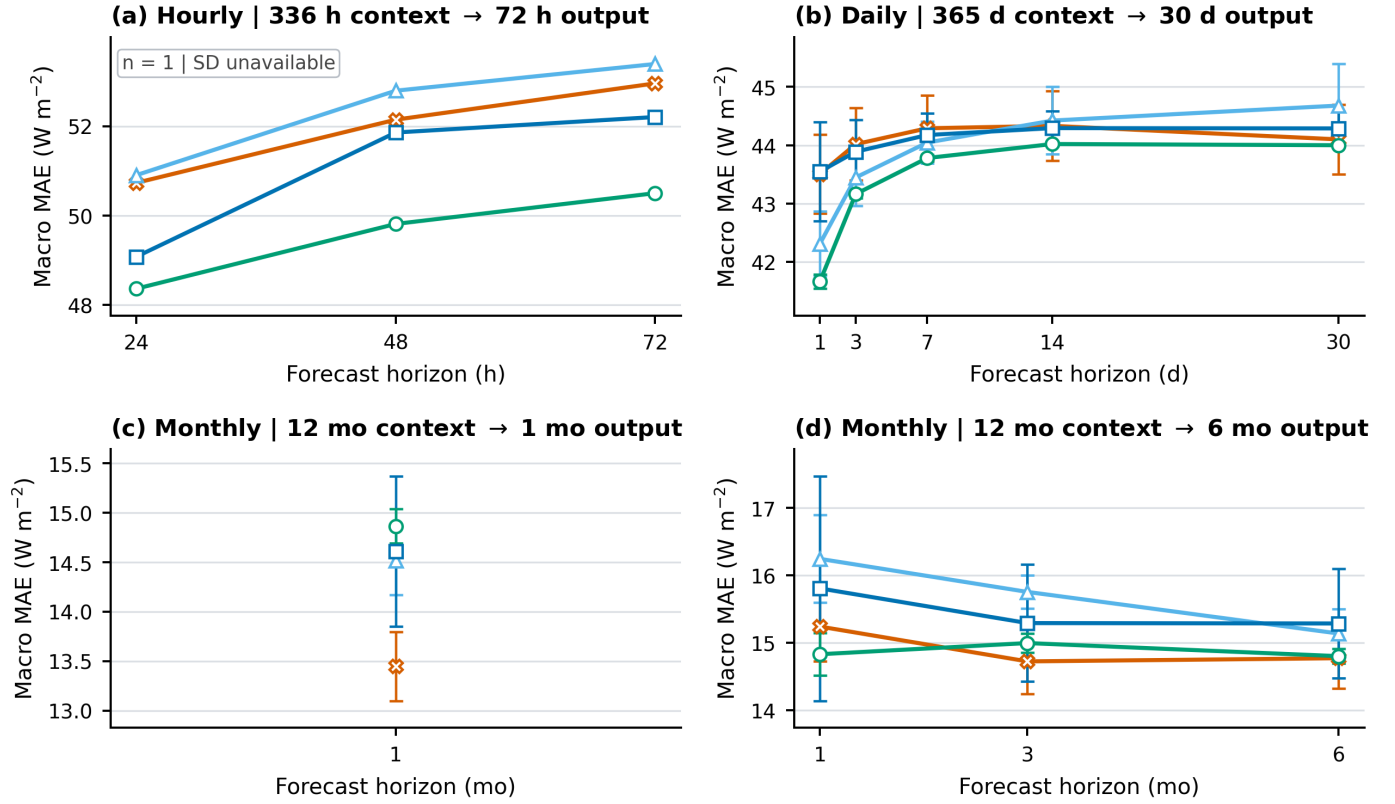


Fig. 6. Mean seed-level macro-MAE with error bars spanning one standard deviation. Marker shape and color jointly identify each model. The hourly extension is plotted as $n = 1$ without an error bar and must not be interpreted as a variability estimate.

adds DilatedRNN with the single official seed 42 to read-only comparator forecasts, whereas daily and monthly estimates use

five seeds. Cross-resolution claims therefore concern protocol behaviour and are not interpreted as equal-precision estimates

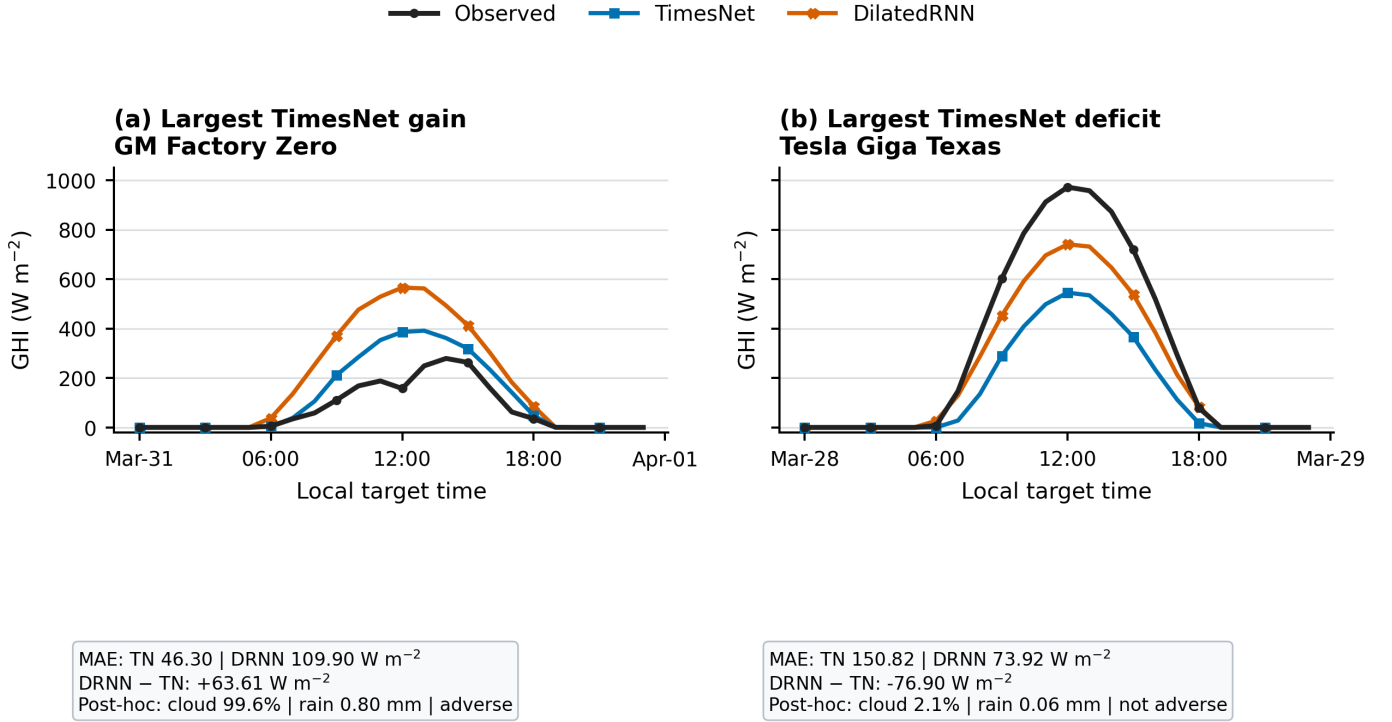


Fig. 7. Observed GHI and model forecasts for the globally selected largest TimesNet gain and deficit. Panels use a shared GHI scale; line style, marker, and color jointly identify each series. MAEs and weather descriptors are annotated in-panel, and meteorology was joined only after case selection.

TABLE XII

METEOROLOGICAL CASE SELECTED INDEPENDENTLY OF MODEL ERRORS

Field	Value
Site	Hyundai Metaplant Georgia
Local date	2024-08-05
Corrected precipitation (mm/day)	82.66
Cloud amount (%)	99.47
Relevant precipitation	Yes
Very cloudy	Yes
Independent adverse condition	Yes
Interpretation	Descriptive meteorological example selected independently of errors.

of random-seed variability.

The neural parameter budgets also differ by task, as reported in Table VIII. Equal optimization rules do not imply equal representational capacity, and the comparison cannot isolate whether a result comes from recurrence, periodic convolution, parameter count, or their interaction. Training and inference runtimes were not instrumented consistently in the completed runs, so no retrospective hardware-efficiency ranking is claimed.

The zero floor imposes a defensible physical bound on all resolutions. The additional hourly nighttime mask uses deterministic solar geometry, but no common upper physical envelope is imposed. Consequently, implausibly high daytime extrapolations remain visible in the errors rather than being hidden by post-processing.

Hourly metrics retain all target hours after the common

nighttime mask. Exact nighttime zeros are part of the operational 72-hour trajectory, but they also make the all-hour error smaller than a daylight-only error would be. The reported hourly values must therefore be interpreted as full-trajectory metrics and not as estimates restricted to irradiance-producing hours.

4) *Contextual-data validity*: The Köppen–Geiger labels are categorical cells from the present nominal 1-km map [46]; they do not resolve microclimates within a cell or conditions on a specific roof. Class boundaries are uncertain, as made explicit by the 50% confidence of the BMW San Luis Potosí assignment. Climate labels are descriptive strata, not model inputs and not explanations of individual forecast errors.

NASA POWER is likewise a gridded, model-based external product rather than local rain-gauge or ceilometer evidence [48]. The 1 mm/day precipitation and 80% cloud thresholds provide reproducible post-hoc tags, but thresholding cannot establish that weather caused a particular error. Weather-case observations therefore supplement the full aggregate evaluation; they do not replace it or determine which model is reported.

VI. CONCLUSION

This study completed a common, leakage-controlled evaluation of TimesNet and DilatedRNN across hourly, daily, and monthly GHI forecasting tasks at ten electric-vehicle manufacturing locations. The matched design fixes inputs, chronological partitions, forecast origins, post-processing, metrics, and location weights within each task, while retaining the different meanings of hourly, daily, and monthly targets.

No architecture was universally superior. TimesNet had the lower macro-MAE at all hourly checkpoints and through 14 days in the daily task. DilatedRNN led at 30 days, in the principal one-month task, and at the three- and six-month prefixes of the exploratory task. Terminal location wins favored DilatedRNN at six of ten hourly, daily, and one-month sites and seven of ten six-month sites, even when a few larger TimesNet gains reversed a macro average. XGBoost led the full benchmark set in the daily and hourly tasks, whereas calendar climatology led the monthly tasks; consequently, the neural comparison does not imply benchmark-wide dominance.

Five-seed daily and monthly results support the observed ranking patterns, but the hourly extension has only the official seed 42 and provides no seed-level uncertainty estimate. Overlapping multi-step targets further restrict the paired evidence to descriptive site-level comparisons. The globally strongest TimesNet gain and deficit both occurred in 24-hour forecasts and had opposite weather contexts, reinforcing that post-hoc meteorology cannot be treated as a causal explanation.

Several terminal architecture differences were small relative to the observed seed variability, and the neural models did not share equal parameter budgets. The evidence therefore supports resolution- and horizon-specific behaviour, not an intrinsic ranking of recurrence against periodic convolution. This interpretation is consistent with the stronger XGBoost and climatology results in their respective tasks.

Future work should evaluate prospective years and unseen sites, repeat the hourly extension over multiple seeds, report daylight-only hourly metrics, instrument training and inference cost, and test whether causal exogenous inputs such as numerical weather prediction and cloud-motion forecasts improve on the strong tabular and climatological references. The released contracts, forecast-origin outputs, seed-level metrics, context manifests, and SHA-256 artifact manifest provide the basis for those extensions.

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DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

DATA AND CODE AVAILABILITY

The source code, experiment contracts, derived evaluation outputs, and integrity manifests used to construct the tables and figures are maintained at <https://github.com/enggusttavomn/TC/tree/codex/finalizacao-artigo-unificado>. External NSRDB, Köppen-Geiger, and NASA POWER products remain subject to their respective providers' access and licensing conditions.

APPENDIX A REPRODUCIBILITY SPECIFICATION

This appendix records the complete configuration needed to reconstruct the matched experiments without inferring values from narrative descriptions. The executable protocol binds task definitions, input and source-code hashes, software versions, model settings, and random seeds into a contract hash. Cached work is reused only when that complete hash matches.

A. Complete temporal contract

Table XIII gives inclusive forecast-origin boundaries. The target cutoff is checked separately, so a multi-step window is retained only when every target lies on or before that date.

The cutoff column records the calendar boundary enforced by the window constructor, not necessarily the last observed target timestamp. In particular, the hourly test's last origin is 2024-12-28 00:00 and its 72 hourly targets end at 2024-12-30 23:00 in the fixed local-time index; UTC timestamps for western sites may extend into 31 December.

The 2019 observations preceding the first daily and monthly origins belong only to the causal context. Selection scales use all available observations before 2023-01-01, and refit scales use all available observations before 2024-01-01. Each scale is site-specific and records its observation count, minimum, maximum, and effective range.

These partitions are rolling-origin evaluations. Parameters and scale values are frozen before the relevant evaluation year, while the context advances and can contain earlier observations from that year after they become available. No parameter update occurs within validation or test.

B. Complete model and optimization settings

The common learned-model input is an array of shape (batch, context length) containing only scaled GHI, accompanied by one integer site identifier. The target is a continuous direct vector of the declared output length. No learned model receives latitude, longitude, clock variables, solar elevation, climate, cloud, precipitation, or a previously predicted target.

Hourly daily-profile persistence repeats the last 24 observed local hours cyclically over the output, and its annual reference preserves the same local hour from the preceding year. For daily/monthly data, persistence repeats the last observed aggregate over the full output. For daily/monthly data, annual seasonal naïve uses the same civil day or month from the preceding year, with 29 February mapped to 28 February when needed. Daily/monthly climatology is the fixed mean for the target calendar day or month using values before the relevant selection or test cut; a global pre-cut mean is the fallback for a missing calendar key. Baselines and learned forecasts receive a zero floor. Only hourly outputs additionally receive the nonpositive-solar-elevation mask, evaluated at minute 30 of each interval.

C. Evaluation and artifact contract

For every declared horizon, the pipeline writes cumulative-prefix and exact-lead metrics. It stores raw and post-processed

TABLE XIII
COMPLETE CHRONOLOGICAL PARTITIONS AND EXPECTED ORIGINS PER SITE.

Task	Partition	Origin range	Declared target cutoff	N
Hourly 336–72	Adjustment	2019-01-15–2022-12-29	2022-12-31	1,445
	Validation	2023-01-01–2023-12-29	2023-12-31	363
	Refit	2019-01-15–2023-12-29	2023-12-31	1,810
	Test	2024-01-01–2024-12-28	2024-12-31	363
Daily 365–30	Adjustment	2020-01-01–2022-12-02	2022-12-31	1,067
	Validation	2023-01-01–2023-12-02	2023-12-31	336
	Refit	2020-01-01–2023-12-02	2023-12-31	1,432
	Test	2024-01-01–2024-12-02	2024-12-31	337
Monthly 12–1	Adjustment	2020-01-31–2022-12-31	2022-12-31	36
	Validation	2023-01-31–2023-12-31	2023-12-31	12
	Refit	2020-01-31–2023-12-31	2023-12-31	48
	Test	2024-01-31–2024-12-31	2024-12-31	12
Monthly 12–6	Adjustment	2020-01-31–2022-07-31	2022-12-31	31
	Validation	2023-01-31–2023-07-31	2023-12-31	7
	Refit	2020-01-31–2023-07-31	2023-12-31	43
	Test	2024-01-31–2024-07-31	2024-12-31	7

TABLE XIV
COMPLETE FROZEN LEARNED-MODEL SETTINGS.

Item	Value
Seeds	11, 23, 42, 67, 89 for daily/monthly; 42 for the official hourly protocol and its DilatedRNN extension
Neural objective	Mean squared error on site-wise min–max normalized GHI
Optimizer	Adam; learning rate 10^{-3} ; weight decay 10^{-5}
Batch/selection	Hourly and daily: batch 128, 30 epochs maximum, patience 5; monthly: batch 32, 300 epochs maximum, patience 30
Determinism	Python, NumPy, and PyTorch generators fixed; deterministic PyTorch algorithms requested; one PyTorch CPU thread
Site representation	Learned embedding of width 8 in TimesNet, DilatedRNN, and LSTM; one-hot vector for XGBoost
TimesNet	$d_{\text{model}} = 8$, $d_{\text{ff}} = 16$, one TimesBlock, top- $k = 3$, two Inception kernels, dropout 0.1, direct linear temporal head; sample-specific FFT period selection and per-window standardization with $\epsilon = 10^{-5}$
DilatedRNN	Vanilla tanh-RNN cells with recurrent dilations (1, 2, 4), 16 units in each layer, dense width 16, dropout 0, direct output head
LSTM	One recurrent layer, hidden width 32, direct output head
XGBoost	Squared-error objective, 150 trees, depth 3, learning rate 0.05, subsample 0.8, column subsample 0.8, histogram tree method, one thread; multi-output-tree strategy when the output length exceeds one
Refit rule	Fresh initialization under the same seed and exactly the epoch count selected on 2023; XGBoost is independently fitted with its fixed budget
Ensemble	Arithmetic mean of five seed forecasts for daily/monthly learned models; no hourly ensemble

forecasts, metrics by site, equal-weight macro metrics, site-paired TimesNet–DilatedRNN differences, selected epoch counts, training histories, and the two sets of pre-cut scales. Daily and monthly task directories additionally store variability across the five seeds. The principal machine-readable files

across task directories are:

- `contrato_execucao.json` and `hiperparametros_e_metodo.json`;
- `protocolo_temporal.csv`, `escalas_minmax_pre_corte.csv`, and `epocas_selecionadas.csv`;
- `previsoes_validacao.csv.gz` and `previsoes_teste.csv.gz`;
- `metricas_por_localidade.csv`, `metricas_macro.csv`, and `variabilidade_sementes.csv` (daily/monthly only);
- `comparacao_pareada_localidades.csv`, `comparacao_pareada_resumo.csv`, and `manifesto_artefatos.json`.

The contract records Python, NumPy, pandas, PyTorch, scikit-learn, XGBoost, and joblib versions together with SHA-256 hashes of source code and input files. Smoke mode reduces sites, origins, seeds, epochs, and trees and is marked nonpublishable in every status artifact.

D. Geographical and meteorological provenance

The geographical CSV retains all factory-coordinate provenance, raster-cell row and column, cell-centre coordinates, sampling distance, class code and descriptions, confidence, source citation, file paths, and the MD5 values in Table XV. MD5 is used only as a reproducibility checksum, not as a security mechanism.

The climate source is the present nominal 1-km Köppen–Geiger product of Beck et al. [46]. The meteorological artifact contains 3,660 location–day rows in local solar time and is accompanied by a JSON manifest that records the NASA POWER query URL and SHA-256 of every cached response [48]. Its declared use is post-hoc error context, with rain at 1 mm/day or more, very cloudy conditions at 80% cloud amount or more, and the logical union as the adverse-day flag. The manifest explicitly records that these variables are not model inputs.

TABLE XV
INTEGRITY RECORDS FOR THE EXTERNAL CONTEXTUAL LAYER.

Asset	Role and retained checksum
Beck present 1-km class raster	Site assignment; MD5 fe8f23532bfb59231a072398f0d9a7cb
Beck present 1-km confidence raster	Assignment confidence; MD5 a87f331e406e4ef02114e1350eb2dd01
Beck present 0.5-degree raster	Americas-map background only; MD5 78d09475fa4b0dbb004b6178cba47b4a
Beck class legend	Codes, descriptions, and RGB colours; MD5 f0632calb3b475d3de5a143c4ee1928f
contexto_geografico.csv	Auditable sampled table; SHA-256 80b6e003a326d6142df03766c15a5dbe28c7c1dc0be3c195df07a60b764f7f41
contexto_meteorologico_2024.csv	Post-hoc NASA POWER context; SHA-256 7fbf97076193f2467e0092cf6b0d2c9a00ea19c84f354b28f1736f5bba95f4b5

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